

LaBSE Fine-tuned Models — Evaluation Report

This report presents the evaluation of 13 fine-tuned LaBSE variants on the IN22-Conv benchmark, compared against the original LaBSE baseline. All models were fine-tuned on specific Indic-Indic language pairs from IN22-Gen and evaluated on the unseen IN22-Conv dataset (1,503 sentences, 462 directed language pairs across all 22 scheduled Indic languages).

The 13 models cover four fine-tuning strategies:

- Single top pair (5 models): fine-tuned on one high-performing language pair each
- Single bottom pair (5 models): fine-tuned on one low-performing language pair each
- All top 5 combined (1 model): fine-tuned on all 5 top pairs together
- All bottom 5 combined (1 model): fine-tuned on all 5 bottom pairs together
- Mixed top5+bottom5 (1 model): fine-tuned on all 10 pairs together

2. Fine-tuning Setup

All 13 models were fine-tuned from the same base model (sentence-transformers/LaBSE) using identical hyperparameters. The only difference between runs is the language pair(s) used for training.

2.1 Hyperparameters

Parameter	Value
Base model	sentence-transformers/LaBSE
Max sequence length	128 tokens
Batch size	32
Epochs	5
Learning rate	5e-6
Warmup ratio	0.1 (10% of total steps)
Warmup steps	14
Steps per epoch	28
Total steps	140
Training examples	900 per run
Validation examples	100 per run
Train/val split	90% train / 10% validation

2.2 Training Data

Each model was trained on sentence pairs drawn from the IN22-Gen dataset. For single-pair models, all 1,000 sentence pairs from the specific directed language pair were used (900 train, 100 validation). For combined and mixed models, pairs from multiple language combinations were pooled before splitting.

The training source is the same IN22-Gen dataset used in the baseline evaluation — meaning the evaluation on IN22-Conv tests genuine generalisation to an unseen conversational domain.

2.3 Model Selection

The best model checkpoint was selected based on validation cosine gap (`val_cosine_gap`) — the difference between mean cosine similarity of true translation pairs and random negative pairs on the held-out 100 validation sentences. This directly optimises for the same discrimination metric reported in evaluation.

3. Evaluation Results — All Models

Sorted by cosine gap (best to worst). LaBSE baseline highlighted in blue.

Model	Cosine Gap	Sensitivity	Specificity	Accuracy@1
bottom_single_5_ben_to_mni	0.4624	0.8867	0.9320	0.5796
bottom_single_4_kan_to_sat	0.4573	0.8842	0.9321	0.5788
top_single_5_guj_to_urd	0.4553	0.8336	0.9257	0.5741
top_single_2_urd_to_guj	0.4549	0.8331	0.9265	0.5757
top_single_4_guj_to_mar	0.4534	0.8314	0.9261	0.5748
top_single_3_ben_to_kan	0.4506	0.8313	0.9260	0.5728
top_single_1_kan_to_tel	0.4401	0.8260	0.9265	0.5730
bottom_single_3_sat_to_hin	0.4232	0.8675	0.9353	0.5855
bottom_single_2_sat_to_urd	0.4199	0.8668	0.9361	0.5853
bottom_single_1_sat_to_npi	0.4135	0.8636	0.9360	0.5837
top5_all_directed_pairs	0.4007	0.8192	0.9272	0.5763
labse_baseline	0.3670	0.8069	0.9180	0.5654
mixed_top5_bottom5	0.3412	0.8929	0.8592	0.4916
bottom5_all_directed_pairs	0.2286	0.8893	0.7750	0.3945

4. Improvement vs LaBSE Baseline

Green = improvement. Red = regression.

Model	Δ Cosine Gap	Δ Sensitivity	Δ Specificity	Δ Accuracy@1
bottom_single_5_ben_to_mni	+0.0954	+0.0799	+0.0141	+0.0142
bottom_single_4_kan_to_sat	+0.0903	+0.0773	+0.0141	+0.0134
top_single_5_guj_to_urd	+0.0883	+0.0267	+0.0077	+0.0087
top_single_2_urd_to_guj	+0.0879	+0.0262	+0.0085	+0.0103
top_single_4_guj_to_mar	+0.0865	+0.0246	+0.0081	+0.0094
top_single_3_ben_to_kan	+0.0836	+0.0244	+0.0080	+0.0074
top_single_1_kan_to_tel	+0.0731	+0.0191	+0.0085	+0.0076
bottom_single_3_sat_to_hin	+0.0562	+0.0607	+0.0174	+0.0201
bottom_single_2_sat_to_urd	+0.0530	+0.0599	+0.0181	+0.0199
bottom_single_1_sat_to_npi	+0.0465	+0.0568	+0.0180	+0.0183
top5_all_directed_pairs	+0.0338	+0.0124	+0.0092	+0.0109
mixed_top5_bottom5	-0.0257	+0.0861	-0.0588	-0.0738
bottom5_all_directed_pairs	-0.1384	+0.0824	-0.1429	-0.1709

5. Key Observations

5.1 Single-pair fine-tuning is the most effective strategy

All 10 single-pair models improve over the LaBSE baseline. The best model is `bottom_single_5_ben_to_mni` with cosine_gap 0.4624 — a +0.0954 improvement over baseline (0.3670), a 26% relative gain. Notably, only 900 training sentences and 140 total training steps were needed to achieve this improvement.

5.2 Bottom-pair models outperform top-pair models

Models fine-tuned on weak language pairs (Santali, Manipuri/Meitei pairings) outperform models fine-tuned on strong pairs. The top 2 models are both bottom-pair models. Fine-tuning on hard, underrepresented scripts forces broader restructuring of the embedding space, producing positive generalisation across all 462 pairs. Top-pair fine-tuning reinforces already-strong patterns with smaller overall benefit.

5.3 Combined training -

`bottom5_all_directed_pairs` is the worst performer — cosine_gap 0.2286 (-0.1384 vs baseline), accuracy@1 drops from 0.5654 to 0.3945. Training on 5 conflicting low-resource language pairs simultaneously with only 900 examples and 5 epochs caused the model to lose general

alignment capability. The training signal from different underrepresented scripts likely conflicted, preventing stable convergence.

5.4 Mixed model trades cosine gap for sensitivity

The mixed model shows the highest sensitivity (0.8929) but low specificity (0.8592) and low accuracy@1 (0.4916). The model identifies true pairs at a high rate but also incorrectly scores many random pairs as translations. Mixed training on 10 diverse pairs pulled embeddings into a dense cluster where the model is overly inclusive.

